EXP 3

Write an R program to create and display a 3D array with specified rows, columns, and tables.

AIM:

To write an R program to create and display a 3D array with specified rows, columns, and tables.

PROCEDURE:

Step 1: Open RStudio or R console on your computer.

Step 2: Create a sequence of numbers to be stored in a variable.

Step 3: Use that sequence to create a multi-dimensional array by specifying dimensions such as number of rows, columns, and tables.

Step 4: Store the created array in a variable.

Step 5: Display the array to view all the elements arranged across multiple rows, columns, and layers.

Step 6: Observe how an array in R can store data in more than two dimensions, unlike a matrix.

CODE:

```
data <- 1:24
arr <- array(data, dim = c(3, 4, 2))
cat("3D Array:\n")
print(arr)
```

OUTPUT:

```
3D Array:
, , 1

     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12

, , 2

     [,1] [,2] [,3] [,4]
[1,]   13   16   19   22
[2,]   14   17   20   23
[3,]   15   18   21   24
```